

CLASS: B.Tech. II Year, I – Sem.

SUBJECT: PROBABILITY AND STATISTICS

Branch: AIML & DS

1. Calculate the correlation coefficient of the given data.

x	12	15	18	21	27
y	2	4	6	8	12

2. Find the correlation coefficient (r) for the following data :

x	38	23	40	4	6	6	55	14	6	47
y	8	8	19	1	10	0	15	4	1	14

3. Find the correlation coefficient (r) for the following data :

Marks in maths	75	30	60	80	53	35	15	40	38	48
Marks in statistics	85	45	54	91	58	63	35	43	45	44

4. Find the correlation coefficient (r) for the following data :

Demand	11	12	13	14	15	16	17	18	19	20
Price	30	29	29	25	24	24	24	21	18	15

5. Find the rank correlation for the following data :

x	56	42	72	36	63	47	55	49	38	42	68	60
y	147	125	160	118	149	128	150	145	115	140	152	155

6. Find the rank correlation for the following data :

R(x)	4	1	6	7	5	2	3	9	10	8
R(y)	6	4	3	1	2	5	10	8	9	7

7. Find the rank correlation for the following data :

A	6	4	3	1	2	5	10	8	9	7
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B	5	2	4	6	3	1	7	10	8	9

8. Find the rank correlation for the following data:

x	84	51	91	60	68	62	86	58	53	47
y	8	36	98	25	75	82	90	62	65	39

9. Find the rank correlation for the following data:

x	56	42	72	36	63	47	55	49	38	42	68	60
y	147	125	160	118	149	128	150	145	115	140	152	155

10. Three judges A, B and C in a beauty contest rank the ten competitors in the following order.

A	1	6	5	10	3	2	4	9	7	8
B	3	5	8	4	7	10	2	1	6	9
C	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient to decide which pair of judges have the nearest approach to common tastes in beauty.